

XML CODE

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:orientation="vertical"    android:padding="20dp"
android:layout_width="match_parent"
android:layout_height="match_parent">

    <EditText
android:id="@+id/etBookName"
android:hint="Enter Book Name"
android:layout_width="match_parent"
android:layout_height="wrap_content"/>

    <EditText
android:id="@+id/etAuthorName"
android:hint="Enter Author Name"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:layout_marginTop="10dp"/>

    <Button    android:id="@+id/btnAdd"
android:text="Add Book"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:layout_marginTop="15dp"/>

    <Button    android:id="@+id/btnShow"
android:text="Show Books"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:layout_marginTop="10dp"/>

    <TextView
android:id="@+id/tvResult"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:layout_marginTop="20dp"
android:textSize="16sp"/>

</LinearLayout>
```

Data Class

```
package com.example.ca
```

```
data class Book(    var
bookName: String? = null,    var
authorName: String? = null
)
```

ActivityMain.kt

```
package com.example.ca
```

```
import android.os.Bundle import android.widget.*
import
androidx.appcompat.app.AppCompatActivity
import com.google.firebase.database.*
```

```
class MainActivity : AppCompatActivity() {
```

```
    lateinit var etBook: EditText
    lateinit var etAuthor: EditText
    lateinit var btnAdd: Button    lateinit
var btnShow: Button    lateinit var
tvResult: TextView
```

```
    lateinit var database: DatabaseReference
```

```
    override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)    setContentView(R.layout.activity_main)
```

```
        etBook = findViewById(R.id.etBookName)
etAuthor = findViewById(R.id.etAuthorName)
btnAdd = findViewById(R.id.btnAdd)    btnShow
= findViewById(R.id.btnShow)    tvResult =
findViewById(R.id.tvResult)
```

```
        database = FirebaseDatabase.getInstance().getReference("Books")
```

```
        btnAdd.setOnClickListener {
```

```
            val bookName = etBook.text.toString()
val authorName = etAuthor.text.toString()
```

```
            if (bookName.isEmpty() || authorName.isEmpty()) {
```

```

        Toast.makeText(this, "Please fill all fields", Toast.LENGTH_SHORT).show()
    } else {

        val bookId = database.push().key!!
        val book = Book(bookName, authorName)

        database.child(bookId).setValue(book)
            .addOnSuccessListener {
                Toast.makeText(this, "Book Added Successfully", Toast.LENGTH_SHORT).show()
etBook.text.clear()          etAuthor.text.clear()
            }
            .addOnFailureListener {
                Toast.makeText(this, "Failed to Add", Toast.LENGTH_SHORT).show()
            }
    }
}

btnShow.setOnClickListener {

    database.addValueEventListener(object : ValueEventListener {
        override fun onDataChange(snapshot: DataSnapshot) {

            val builder = StringBuilder()

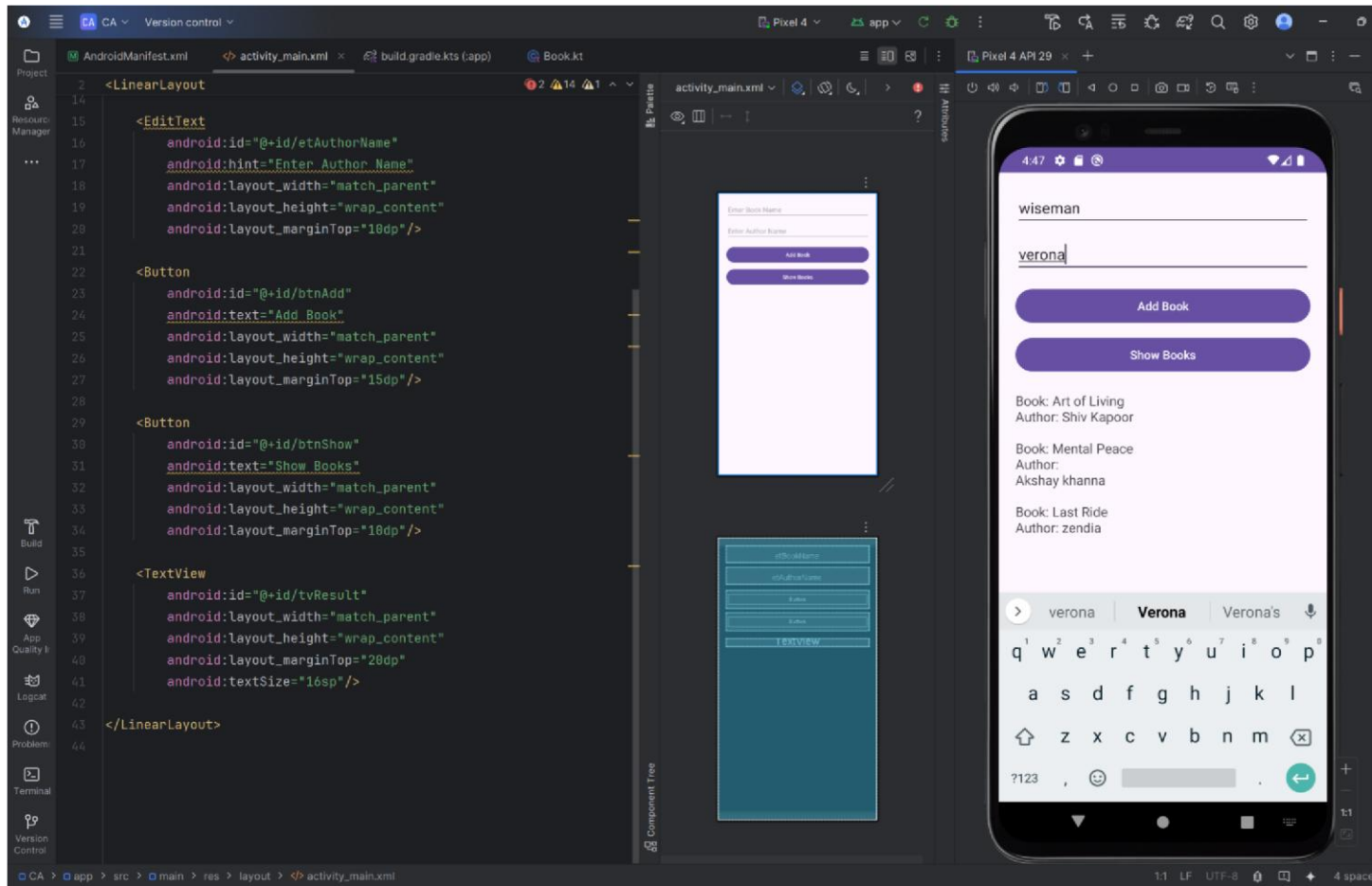
            for (data in snapshot.children) {          val book =
data.getValue(Book::class.java)
            builder.append("Book: ${book?.bookName}\n")
            builder.append("Author: ${book?.authorName}\n\n")
            }

            tvResult.text = builder.toString()
        }

        override fun onCancelled(error: DatabaseError) {
            Toast.makeText(this@MainActivity, "Error", Toast.LENGTH_SHORT).show()
        }
    })
}
}}

```

UI Output Screen



FireBase Realtime Database

The screenshot shows the Firebase Realtime Database console for a project named 'StudentRecords1'. The left sidebar contains navigation links for Project Overview, Authentication, Realtime Database (selected), Phone Verification, and Product categories. The main area displays the Realtime Database interface with tabs for Data, Rules, Backups, Usage, and Extensions. A URL bar shows the database endpoint: `https://studentrecords1-96ebd-default-rtdb.firebaseio.com/`. The data is structured as a JSON tree under the 'Books' node, containing four entries with author and book names. A 'Need help with Realtime Database? Ask Gemini' button is visible. The database location is noted as 'United States (us-central1)'.

StudentRecords1

Realtime Database

Need help with Realtime Database? Ask Gemini

Data Rules Backups Usage Extensions

Protect your Realtime Database resources from abuse, such as billing fraud or phishing Configure App Check

`https://studentrecords1-96ebd-default-rtdb.firebaseio.com/`

```
https://studentrecords1-96ebd-default-rtdb.firebaseio.com/  
  
└ Books  
  └ -01fHQfGYZNdYu73gqE  
    └ authorName: "Shiv Kapoor"  
      └ bookName: "Art of Living"  
  └ -01fIUciSSY-kp-vOKw  
    └ authorName: "Akshay khanna"  
      └ bookName: "Mental Peace"  
  └ -01fJ3T2WLXAuODJ_JZj  
    └ authorName: "zendia"  
      └ bookName: "Last Ride"  
  └ -01fLmemw67oYhP7iYLI  
    └ authorName: "verona"  
      └ bookName: "wiseman"
```

Database location: United States (us-central1)